

**MIT Art Design and Technology University
MIT School of Computing, Pune**



Department of ASH

First Year Engineering

23ASH1104-Applied Sciences (Materials for Computing Devices)

Class - F.Y. (SEM-I)

Unit - II

Engineering Materials

Dr. Rajesh Jadhav

Dr. Manoj Patowary

Dr. Vaishali Inamdar

Dr. Sachin Musale

Prof. Laxmikant Jagtap

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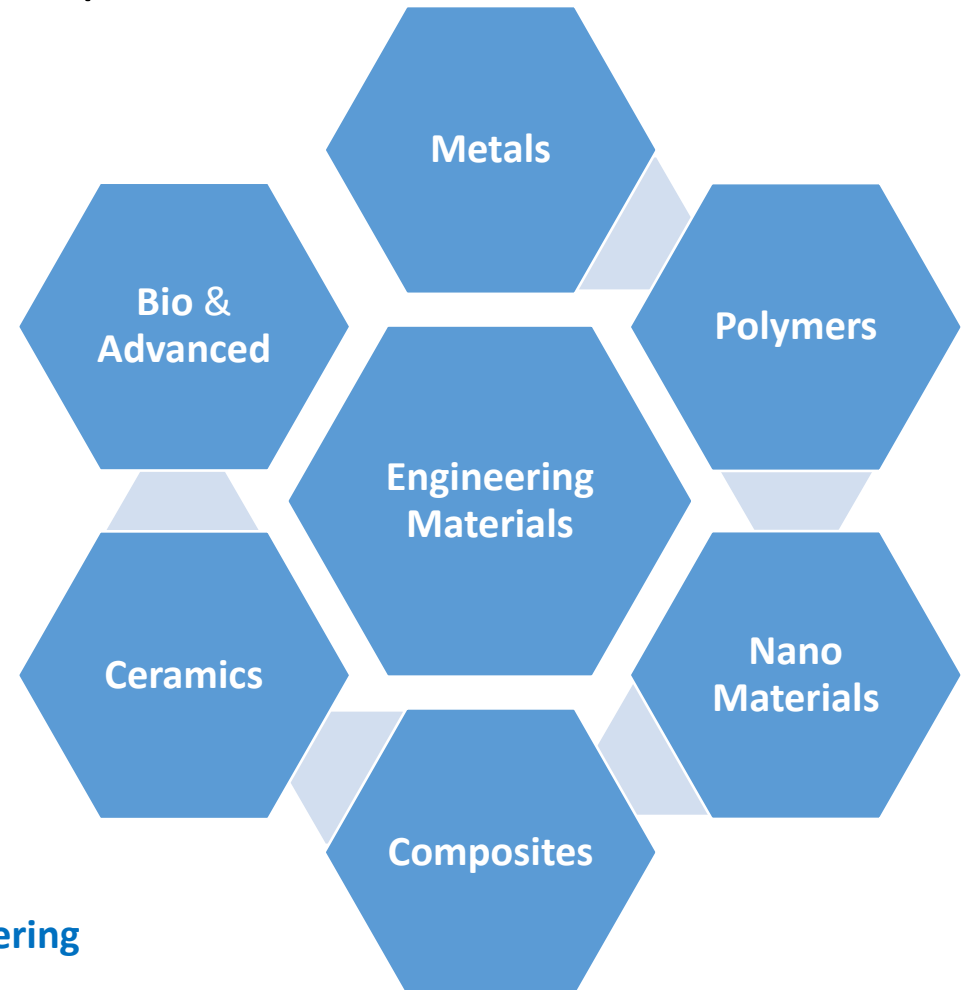


Unit II - Syllabus

- **Main Topic-1:** [A] Materials for Engineers: Introduction, Classification, Structure Property Relationship, Periodicity of properties
-
- **Main Topic-2:** [B] Nano technology and computational development: Introduction, (Categories of Nano Computing, Benefits of nanotechnology for computer science, Role of nanotechnology in computer science,) Nanotechnology and transistor building, Nanotechnology and quantum computers processor building, Nanotechnology and memory building, Optical nano technology.
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- **Main Topic-3:** [C] Role of computer science in nanotechnology Image enhancement with digital signal

Engineering Materials

Engineering Materials is refer to a group of materials used in the construction of man made structure and components.



Engineering Materials

- **Metals**

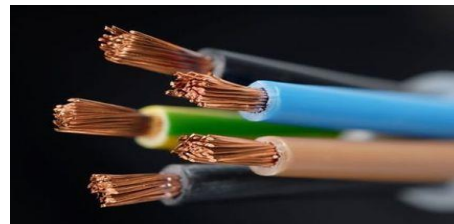
Silver in Computer Chips



Platinum – Hard Drives



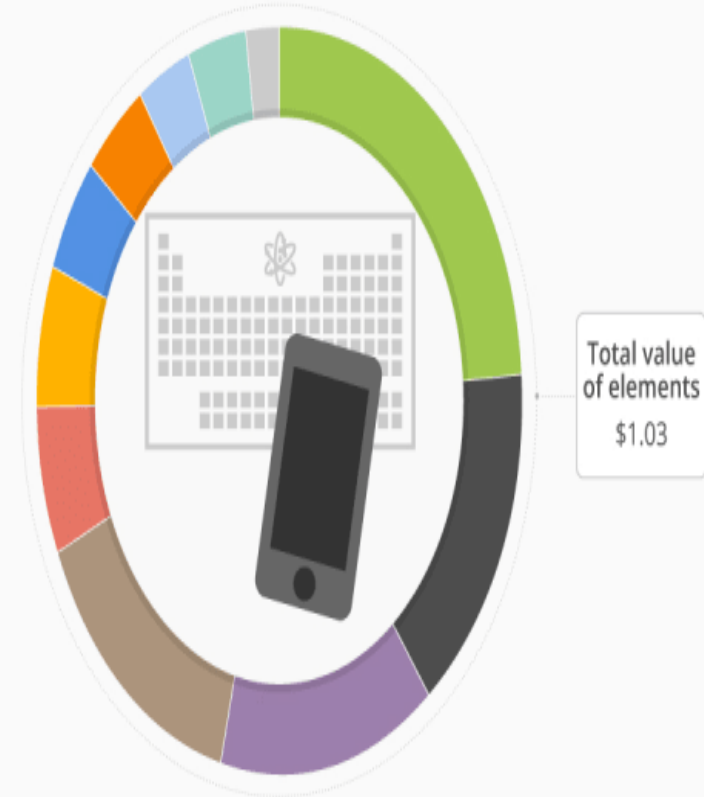
Copper –wiring, and cables



129 Grams: The Materials That Make Up The iPhone

Materials used in iPhone 6, 16GB model

31.1 g	Aluminium
19.9 g	Carbon
18.7 g	Oxygen
18.6 g	Iron
8.1 g	Silicon
7.8 g	Copper
6.6 g	Cobalt
5.5 g	Hydrogen
4.9 g	Chrome
4.9 g	Others
2.7 g	Nickel
129.0 g	Total



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Engineering Materials

- **Polymer-**

1) PPV

2) ABS

3) PVC

4) Polycarbonate

5) Conducting polymer

PPV Used in LED



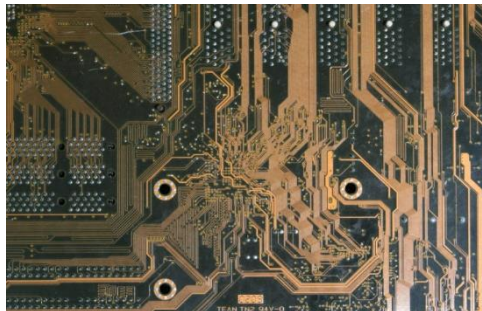
Polycarbonate in laptop bodies



PVC in Cables



ABS in Keyboard



Conducting polymer likes Polyaniline used in IC

Nanomaterial Introduction :

□ Definition:-

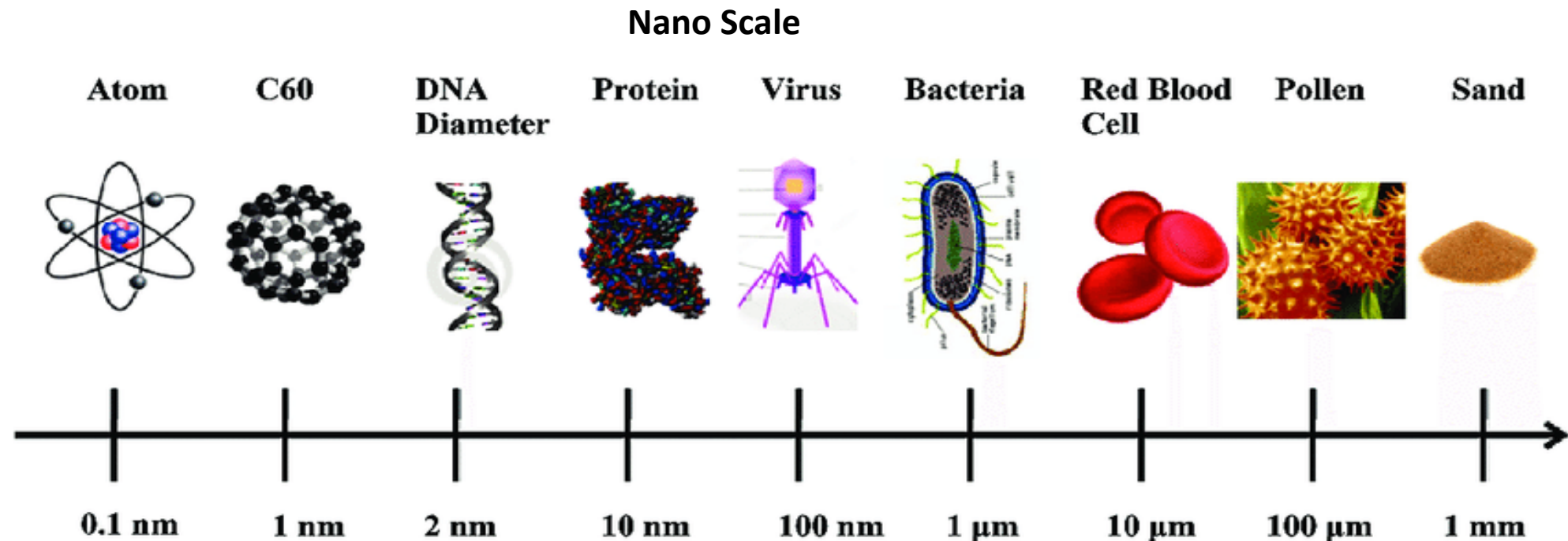
Nanomaterials are defined as a set of substances where at least one dimension is less than 100 nanometres



What is Nano-technology ??

- ❑ Nanotechnology can be defined as the design, characterization, production and application of structures devices and systems by controlling shape and size at a Nano meter Scale.

Professor Norio Taniguchi was the first person to use the term 'nanotechnology' in 1974.



Nanomaterial

▪ why are nanomaterials so special?

the properties of nano materials are very much different from those at a larger scale.

▪ Physical properties change due to :

I) large surface area to volume ratio

i) large fraction of surface atoms

ii) large surface energy

II) quantum confinement

III) reduced imperfections

▪ Types of Physical properties :

Optical (e.g. color, transparency)

Electrical (e.g. conductivity)

Mechanical (e.g. hardness, melting point)

Magnetic (e.g. Super Paramagnetism)

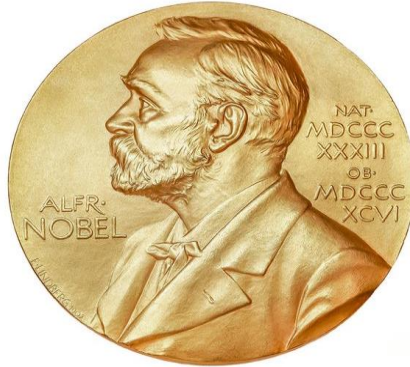
Nano dimensional Structures

□ What are Nano dimensional Structures?

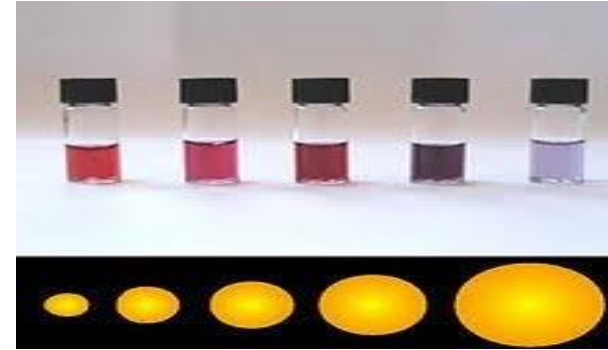
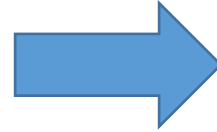
- When one or more of the dimensions of a solid are reduced significantly, its physicochemical characteristics notably depart from those of the bulk solid.
- With reduction in size, novel electrical, mechanical, chemical, magnetic, and optical properties can be introduced. The resulting structure is then called a low dimensional structure (or system).
- Nanostructures constitute a bridge between molecules and bulk materials.

Suitable control of the properties and responses of nanostructures can lead to new devices and technologies.

Bulk Gold to Gold Nanoparticles



Gold in jewellery



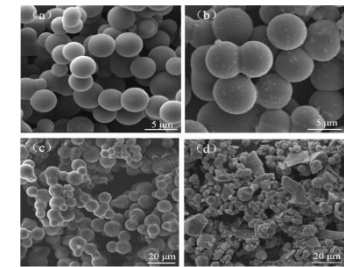
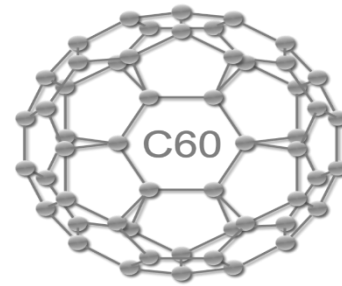
Gold nanoparticles color vary as their size vary

- 1) Bulk Gold appears Yellow in color.
- 2) Nanosized Gold appears Red in color
- 3) The particles are so small that electrons are not free to move about as in bulk gold.

Nanomaterials: Classification

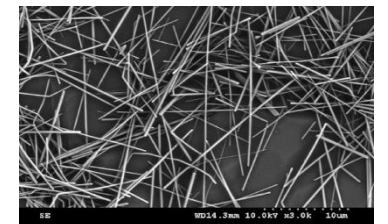
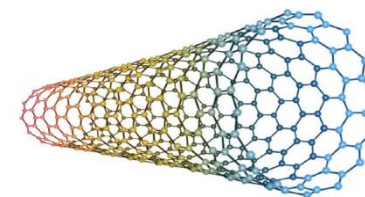
1) **Zero Dimensional (0D)**:- materials having all the dimensions measured within Nano scale.

E.g.: Nano dots, Quantum dots, Fullerenes



2) **One Dimensional (1D)**:- materials having two of three dimensions are measured within Nano scale.

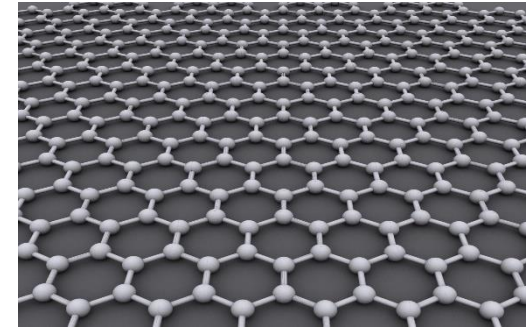
E.g. Nanowires, nanorods, nanotubes, nanopillars, Carbon NanoTubes (CNT)



Nanomaterials: Classification

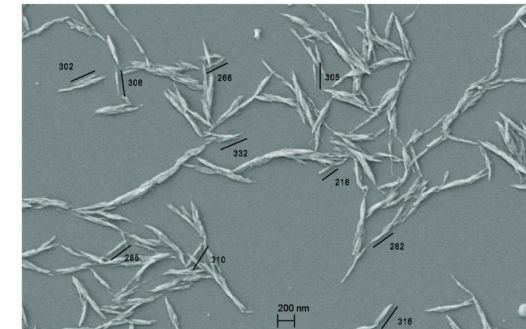
3) **Two Dimensional (2D):-** Materials having only one of three dimensions are measured within Nano scale.

E.g.: Nanofilms, nanocoatings, nanoplates, Graphene,

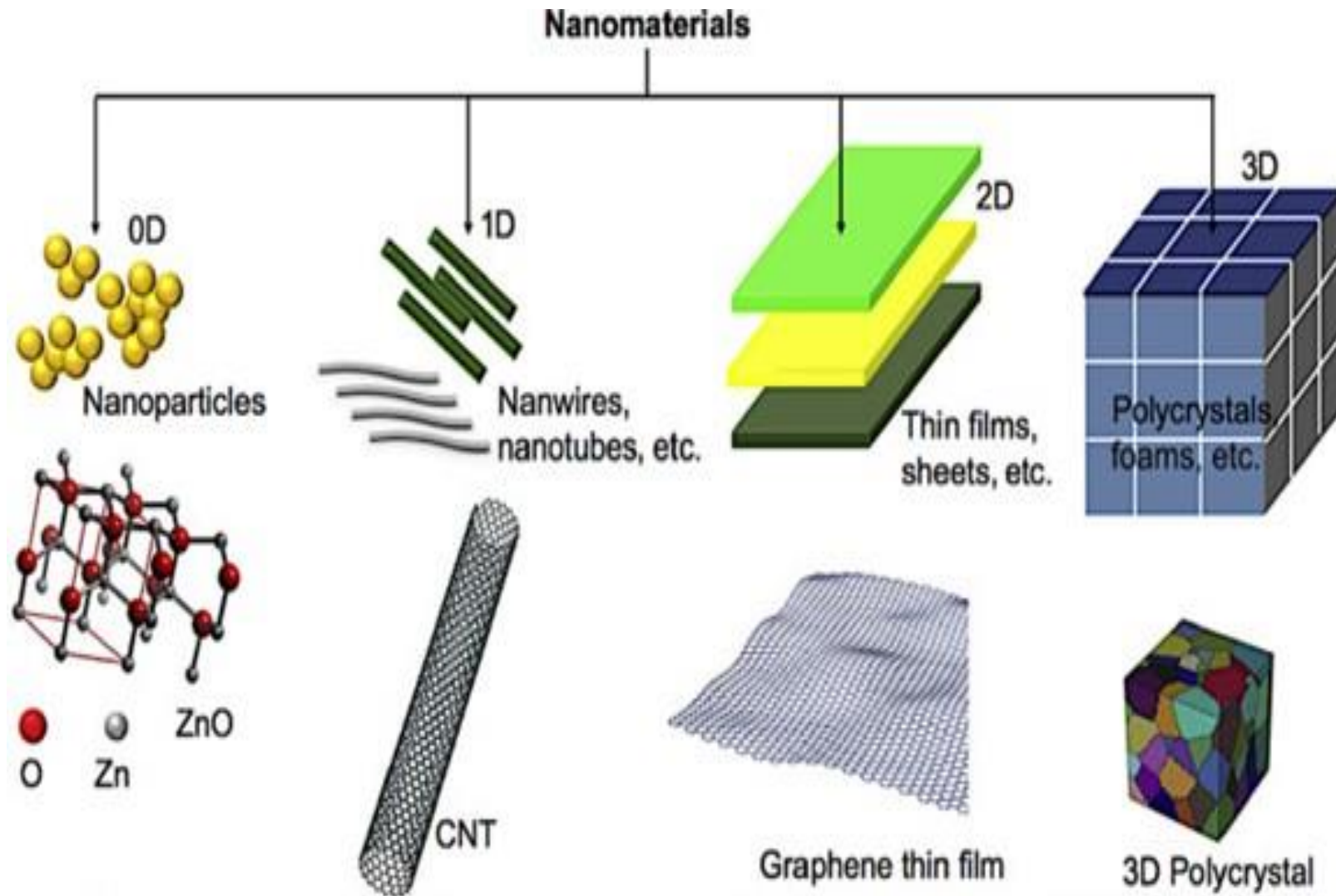


4) **Three Dimensional (3D) or Nano Bulk structure:-** Nano materials having all the three dimensions not are measured within Nano scale.

E.g.: Nano crystal



Nanomaterials: Classification



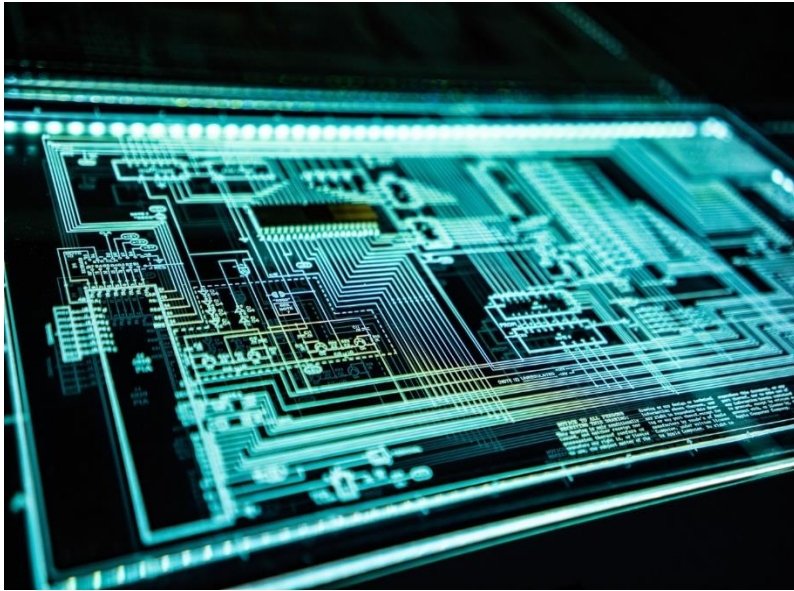
Nano Computing:-

Definition - Nano technology is used in computer science by creating more efficient computing components called as Nano Computing.

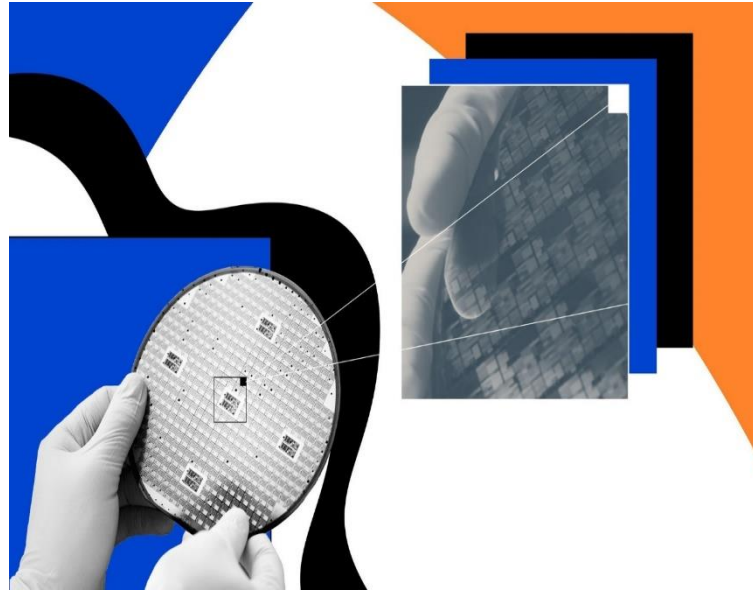


Nano Computing

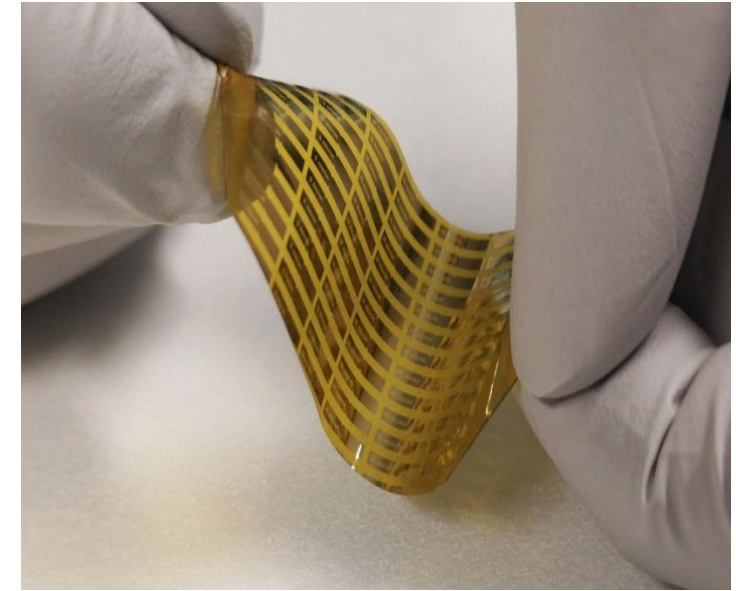
Nano computing is a computer that uses parts and components measured in nanometers.



Nano computers utilize nanoscale components, such as nanowires, nanotubes, or quantum dots, to perform computations and manipulate data



ability to store vast amounts of data in a compact form, making them suitable for applications requiring high-density memory



Due to their small size, nano computers can be integrated into a wide range of applications

Categories of Nano computing

- 1) **Molecular Nano computing**:- utilize individual molecules or groups of molecules as the fundamental building blocks of computation.
- 2) **Quantum Nano computing** :- Quantum Nano computers rely on quantum bits, or qubits, which can exist in multiple states simultaneously.
- 3) **Nano-electromechanical systems (NEMS) computing** :- These systems use tiny mechanical elements, such as nanowires or nanotubes, that can exhibit mechanical motion, such as vibration or bending, in response to electrical signals.
- 4) **DNA-based Nano computing** :- utilize the unique properties of DNA, such as its ability to store and process information through base pairing, to perform computations.

Benefits of Nano for computer science

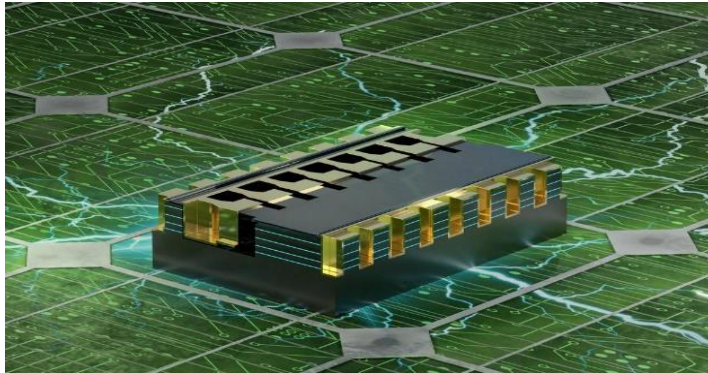
- 1) **Size and portability:-** Their miniature size would allow for extraordinary portability and integration into various objects and systems.
- 2) **Tiny but powerful:-** Nano computers could potentially offer a significant increase in computing power compared to traditional computers.
- 3) **Energy efficient:-** Due to their miniature size, they would require minimal amounts of energy to function, making them highly energy-efficient.
- 4) **Integration into complex systems:-** The small size of nano computers would make them well-suited for integration into complex systems, including biological systems.

Nano technology in Transistor

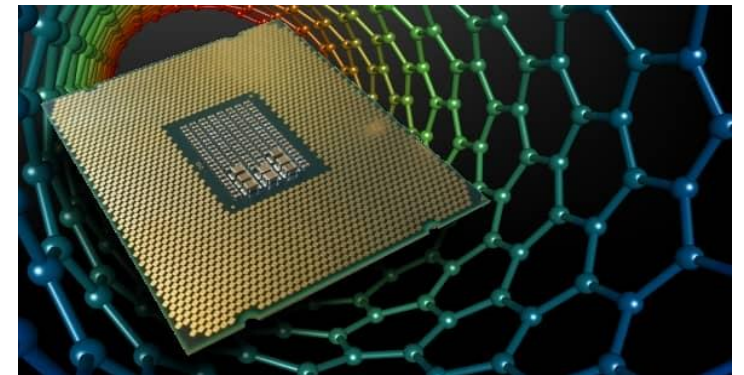
Nano Transistor:- It is a semiconductor device in nano scale that is used to control or regulate the flow of electronic signals.

Why Nano Transistor?

- i) The superior in electrical properties, responsiveness,
- ii) capacity to interface with other systems,
- iii) high stability (including thermal stability), natural tiny size.

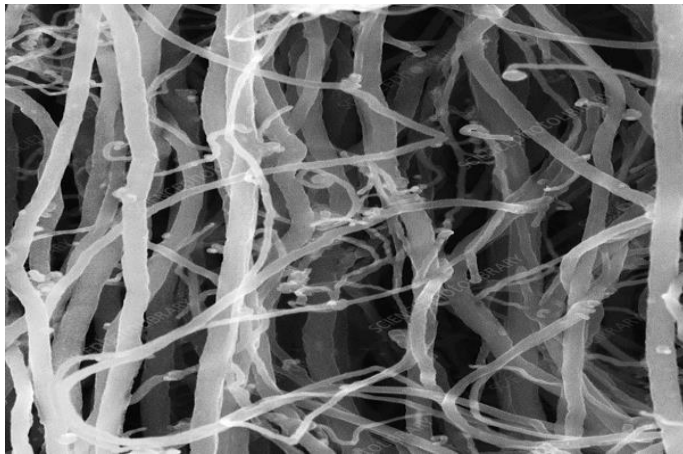
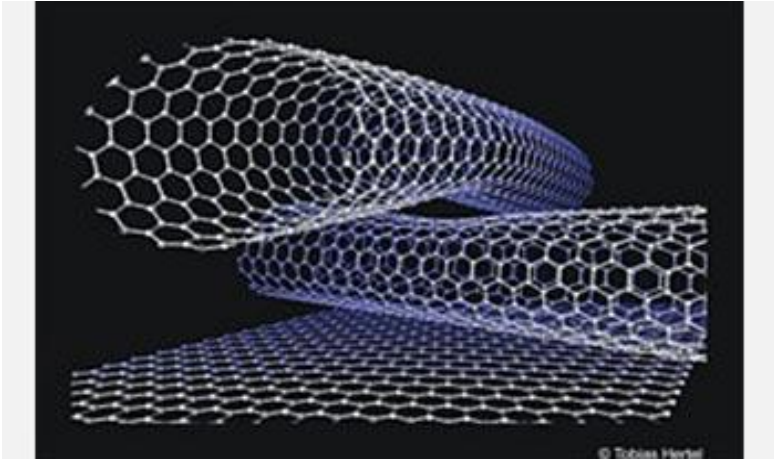


Nano Transistor

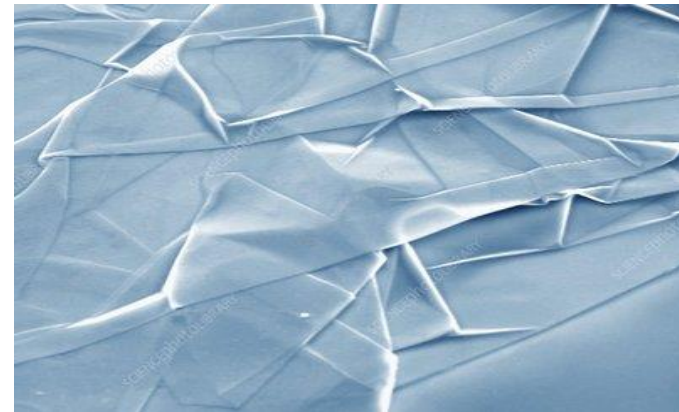
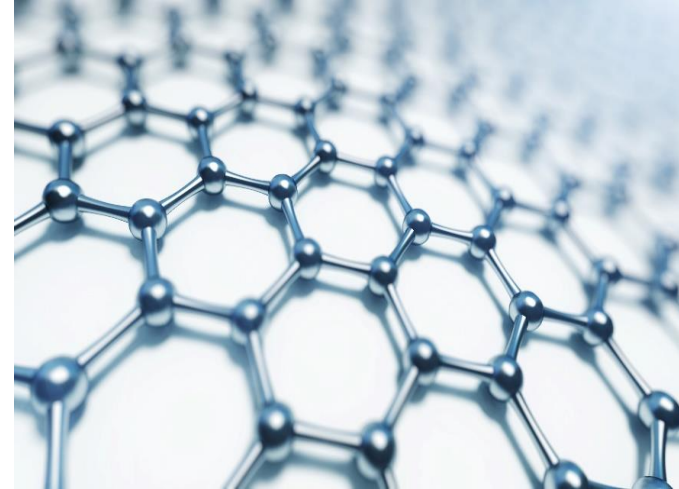


berkeley-creates-a-1-nanometer-gate-transistor-the-smallest-ever

Eg.:- Materials used in Nano transistor:-



carbon nanotubes,

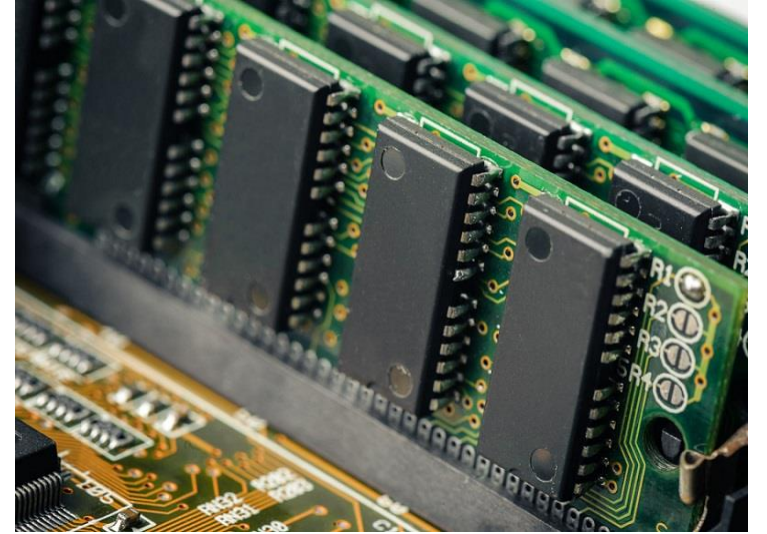


Graphene

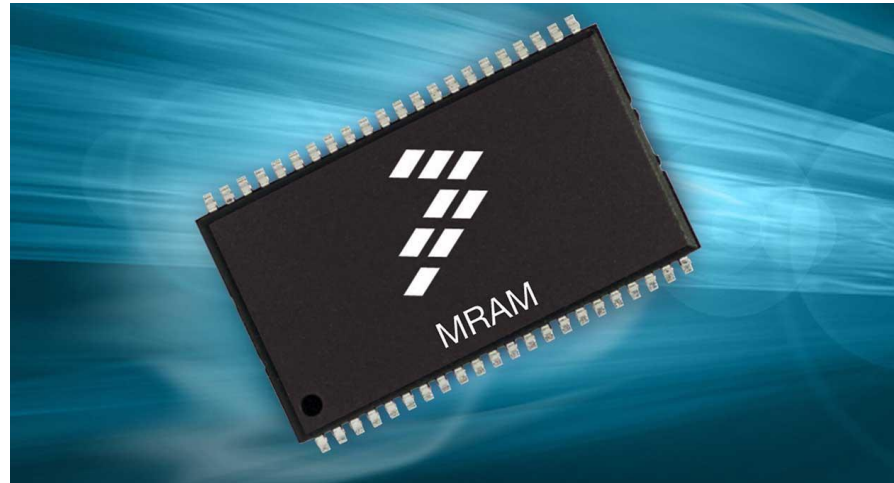
Nano technology in Memory building



Hard disc drive



RAM



MRAM

Nano technology in Memory building

1) **Hard drives**- It is a part of hardware used to store digital content and data on computers. It need more power and possess a higher failure rate.

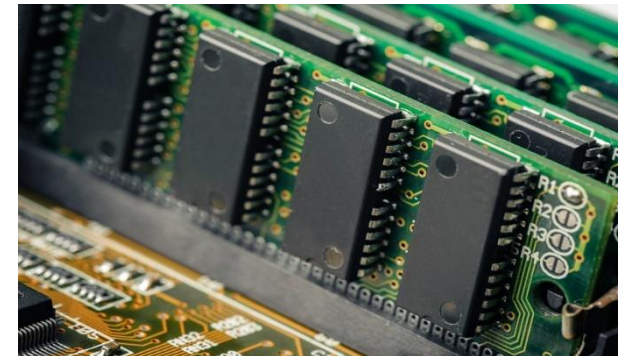
- **Materials used**- Flat circulated plates of Aluminium or glass & coated with magnetic materials.

2) **RAM**- RAM is a temporary memory bank where your computer stores data it needs to retrieve quickly. (RAM) consumes less space, less battery power, and is less likely to be destroyed in the event of a device drop.

- **Materials used**- Made from silicon aluminium & copper



Hard disc drive



RAM

Nano technology in Memory building

3) **MRAM**- Magneto resistive random access memory (MRAM) is a non volatile random access memory which store data in magnetic domain.

- **Importance of MRAM-**

- i) Non volatility
- ii) High speed performance
- iii) Low cost
- iv) Infinite endurance

- **Materials used in MRAM:-** Magnetic nanowires made of an alloy of iron and nickel, nickel iron cobalt, cobalt iron.

- **Applications of MRAM-**

In all memory applications in digital camera, cellular phones, Mp3, laptops
In automotive industry, military & space applications.



First 200mm 1 Mb MRAM wafer, fabricated by Motorola, 2001

Quantum Computing

Definition- quantum computing is the use of quantum mechanics or the description of things at the atomic and subatomic levels to dramatically increase computer processing power and speed.

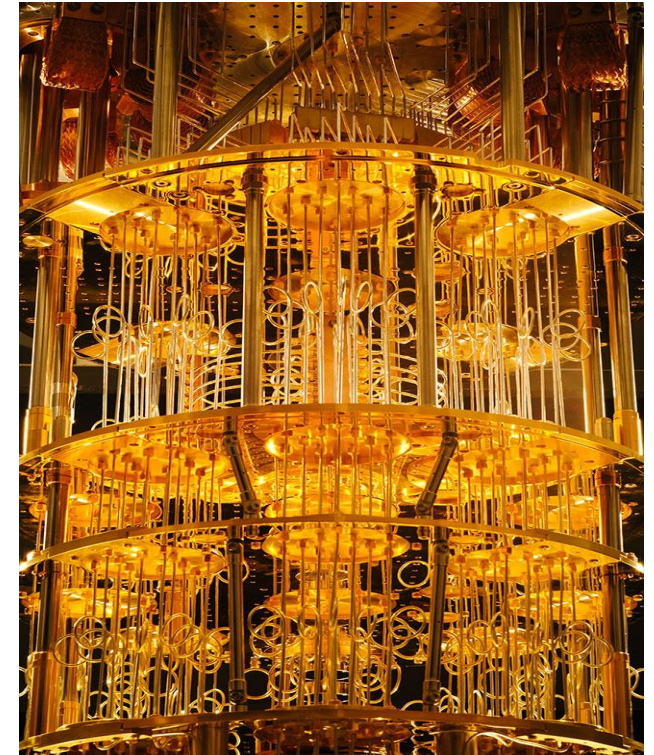


Quantum Computing

- Why Quantum Computing-

- 1) quantum computing's core benefit is the acceleration of computing processes it makes regular computing way more powerful.
- 2) Quantum computing also has incredible importance for the potential of machine learning and artificial intelligence.

Quantum computer



Quantum Computing

- Applications Of quantum computing-

- 1) **Cryptography**- quantum computer have the potential to keep private data safe.
- 2) **In computational chemistry**- quantum computing is solving classically intractable chemistry problems This may help to answer unresolved questions about phenomena like: transition metal catalysis, or certain biochemical reactions.
- 3) **Machine learning**- Research indicates that quantum computing could significantly accelerate Machine learning & data analysis task.
- 4) **Medicine & materials**- quantum computer mimics the computing style of nature, allowing it to simulate, understand & improve upon natural things-like molecule & their interaction.
- 5) It is used in weather forecasting, logistic optimization & in financial modelling.

DNA Computing

In 1994, Leonard Adleman

- **Definition of DNA Computing-** It is a natural computing based on the concept of performing logical and arithmetic operations using molecular properties of DNA by replacing traditional carbon/silicon chips with biochips.



DNA-based nano computers leverage the unique properties of DNA, such as its ability to store and process vast amounts of information in a compact form

DNA COMPUTING

- **Importance/need of DNA computing-**

- 1) conventional or traditional silicon based computers have a limited speed and beyond a point cannot be miniaturize.
- 2) information storage capacity of DNA molecule is much higher than silicon based computers. it has ability to store 10^8 terabytes of data in 1 gram of DNA
- 3) operations on dna computing are parallel, test tube of dna may contain around trillions of strands. each operation are carried out in all the strands parallel.

DNA COMPUTING

Applications Of DNA Computing-

1) Security-

deploying DNA algorithms in cryptography to build an intrusion detection system (ids it is a device or software application that monitors a network or systems for malicious activity or policy violations.)

2) Clustering –

it deals with deriving highly meaningful relationships in a complex collection of data by creating a structure using various concepts and algorithms.

DNA COMPUTING

Drawbacks of DNA COMPUTING-

1) Complex & expensive tools-

the output yielded by DNA computation would require complex expensive tools.

2) Probability of mismatching breaks-

The synthesis of DNA is also prone to errors such as mismatching breaks. The possibility of errors increasing exponentially with each iteration which reduces the reliability.

Optical Nano technology

❑ **Optical components-** It is the object which interacts with light so as to either modify the light or convert light to electricity OR electricity into light.

Examples – Lens, grating, mirror, fiber

❑ **Suitable properties of Nano materials used in optical technology-**

- 1) Nano materials are small in size enables components & devices they are used to be smaller in size.
- 2) The high surface area of nano materials can be exploited to manipulate & change the property of light.
- 3) The abrasion resistance & toughness of many nano materials are also useful which helps to prevent damage of optical components & raise the efficiency of it.

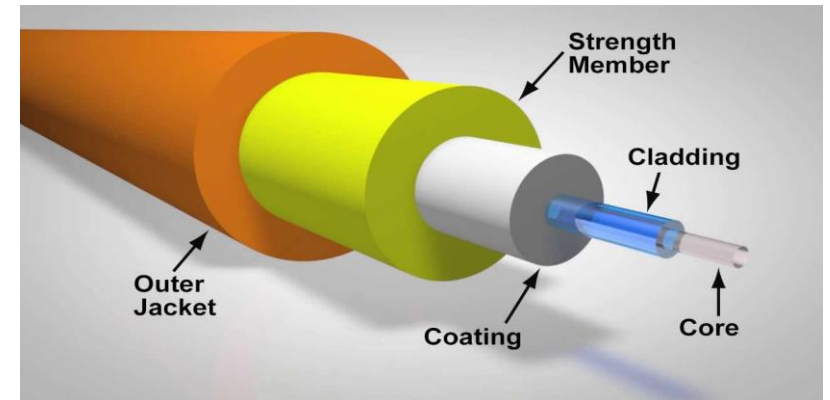
❑ **Nano optics:-**

It is the materials shows interaction of light with their nano particles called as nano optics.

Optical Nano technology

- ❑ **Definition of optical fibre-** It refers to the technology that transmits information as light pulses along a glass or plastic fiber, travelling to distances of hundreds of miles with higher bandwidth.
- ❑ In today's world, optical fiber cables are the primary source of high-bandwidth, long-distance communications between multisite organizations, telephone companies, and several other long-distance communications applications.

Structure of optical fiber



The core, cladding, and outer coating are all components of an **optical fibre**. While glass and plastic are commonly utilized, various materials can be used based on the desired transmission spectrum.

Optical Nano technology

□ Types of optical fibre based on materials-

A) Glass fibre

Example:

Core: SiO₂ Cladding: SiO₂

Core: GeO₂- SiO₂ Cladding: SiO₂

B) Plastic fiber

Example:

Core: Made of Polystyrene : Cladding: Done with Methyl Methacrylate

Core: Made of Polymethyl Methacrylate : Cladding: Done with Co-Polymer

Optical Nano technology

Applications of optical fibre –

1) Communication:

Optical fibre cables play an important role in the principle of optical fibre communication systems because they are used for both transmission and reception.

2) Defence:

High-level data security sectors of military and aerospace applications use fiber optics for data transfer. Hydrophones for SONAR and seismic applications, as well as aviation wiring, make use of this material.

3) Broadcasting:

Such cables deliver high-speed, high-bandwidth HDTV signals. When compared to the same amount of copper cables, fibre optic cable is less expensive.

4) Medical industry:

Surgical lasers, endoscope lasers, microscope lasers, and biomedical lasers all make use of fiber lasers.